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DEV - Project no. 1 Results

Results analysis

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Project outline

Experiment outline

The goal of this experiment is to evaluate the overall quality of selected clustering algorithms on a wide range of benchmark datasets. The testing will focus on:

- **43 2-dimensional datasets**, with sets of reference labels (`label0`)
- **6 clustering algorithms** (K-means, Gaussian mixture model, Genie (with different g parameter values), Agglomerative Hierarchical Clustering (with single, average, complete and Ward linkage), DBSCAN, spectral clustering).

Clustering evaluation

The evaluation of the clustering performance is made on 2 bases:

- internal cluster metrics, such as Silhouette score,
- metrics based on the reference labels, prepared by the authors of the datasets, such as Adjusted Rand Index (ARI).

Experiments

DBSCAN experimentation issues

As said in the assignment description DBSCAN algorithm has to be considered separately, since opposed to other clustering algorithms it estimates the number of clusters on its own. For other algorithms, the number of clusters is passed as input, and the number of requested clusters is equal to number of clusters found in the reference labels.
Additional for DBSCAN parameters ϵ and *MinPts* have to be estimated.

DBSCAN experimentation approach (1)

Parameter estimation:

1. *MinPts* estimation
   - heuristic 1 - $MinPts >= d + 1 => MinPts >= 3$
   - heuristic 2 - $MinPts \in \{2d, 4d\} => MinPts \in \{4, 8\}$

   Therefore for each dataset I tested $MinPts = \{3, 4, 5, 6, 7, 8\}$.
2. ϵ estimation - The optimal ϵ was estimated for each of the *MinPts* values using the *KneeLocator* method.

After choosing the optimal ϵ for each *MinPts* value and evaluating each model using *ARI*, the model maximizing *ARI* was chosen and its partitions were returned.

DBSCAN experimentation approach (2)

Additionally:

- all points marked as noise by the algorithm (with $label = -1$) were removed,
- all label values were increased by 1 to be consistent with the reference labels.

The mismatch of number of predicted clusters with the reference labels was not a problem for internal cluster metrics, ARI and NMI. However for NCA, a direct mapping of labels is needed, therefore this metric was not calculated for DBSCAN. In other cases NCA metrics displayed a high correlation with ARI and NMI, therefore skipping this metric did not compromise the experiment.

Overall experimentation approach

All algorithms were tested on all 43 datasets and their performance has been saved into a dataframe. The dataframe consists of:

- dataset and battery name,
- algorithm name (and the g parameter value for Genie),
- reference metrics values (ARI, NMI, NCA),
- internal metrics values (Silhouette, Calinski Harabasz etc.),
- median of reference metrics,
- median of internal metrics.

Additionally for DBSCAN experiments I included:

- the number of clusters in the reference labels and the number of clusters produced by DBSCAN,
- *MinPts* and ϵ parameters values,
- coverage (the percent of points which have not been classified as noise).

All further analyses have been performed on these data frames.

Results dataframe

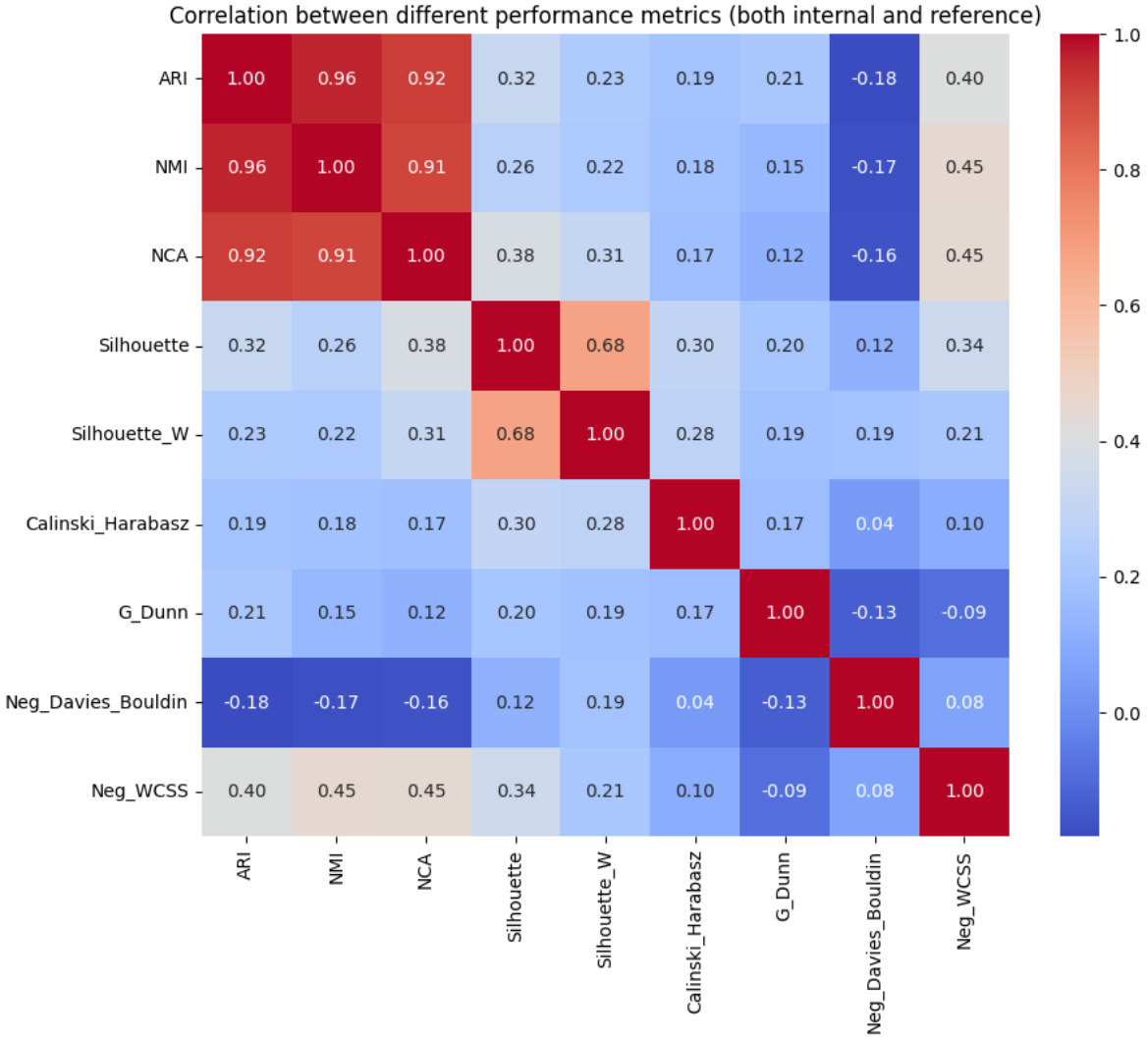
	Dataset	Algorithm	ARI	NMI	NCA	Silhouette	Silhouette_W	Calinski_Harabasz	G_Dunn
0	a1	K-means	0.890841	0.946220	0.907719	0.549520	0.547436	10905.204893	0.034414
1	a1	GMM	0.879836	0.942423	0.903158	0.545172	0.547277	10444.250648	0.040625
2	a1	Genie_g0.1	0.937118	0.958716	0.966667	0.585828	0.586648	13331.892006	0.379052
3	a1	Genie_g0.3	0.898836	0.952200	0.911579	0.563789	0.569258	11325.882022	0.258961
4	a1	Genie_g0.5	0.834178	0.937344	0.840000	0.481258	0.559391	8538.239593	0.301465
5	a1	Genie_g0.7	0.584754	0.865073	0.576491	0.146351	0.466938	3347.135167	0.251492
6	a1	Genie_g0.9	0.443571	0.787220	0.371228	0.013509	0.340636	1570.179112	0.293158
7	a1	Agg_single	0.443571	0.787220	0.371228	0.013509	0.340636	1570.179112	0.293158
8	a1	Agg_average	0.925094	0.951751	0.960000	0.581964	0.582859	13137.096064	0.339183
9	a1	Agg_complete	0.916228	0.950552	0.955439	0.575687	0.580996	12809.890146	0.181331
10	a1	Agg_ward	0.914155	0.947310	0.954386	0.576202	0.580322	12870.148120	0.232850
11	a1	SC	0.811431	0.901356	0.858947	0.511693	0.521961	9077.310102	0.050927
12	a2	K-means	0.927003	0.966046	0.942941	0.578303	0.574400	16482.207779	0.024344
13	a2	GMM	0.927116	0.966989	0.941373	0.578581	0.577424	16398.663386	0.044378

Figure: Dataframe with results from all experiments.

Analysis of results

Correlation of metrics

The reference metrics are very correlated, therefore most analyses were focused on the ARI and Silhouette metric.



Mean ARI per dataset

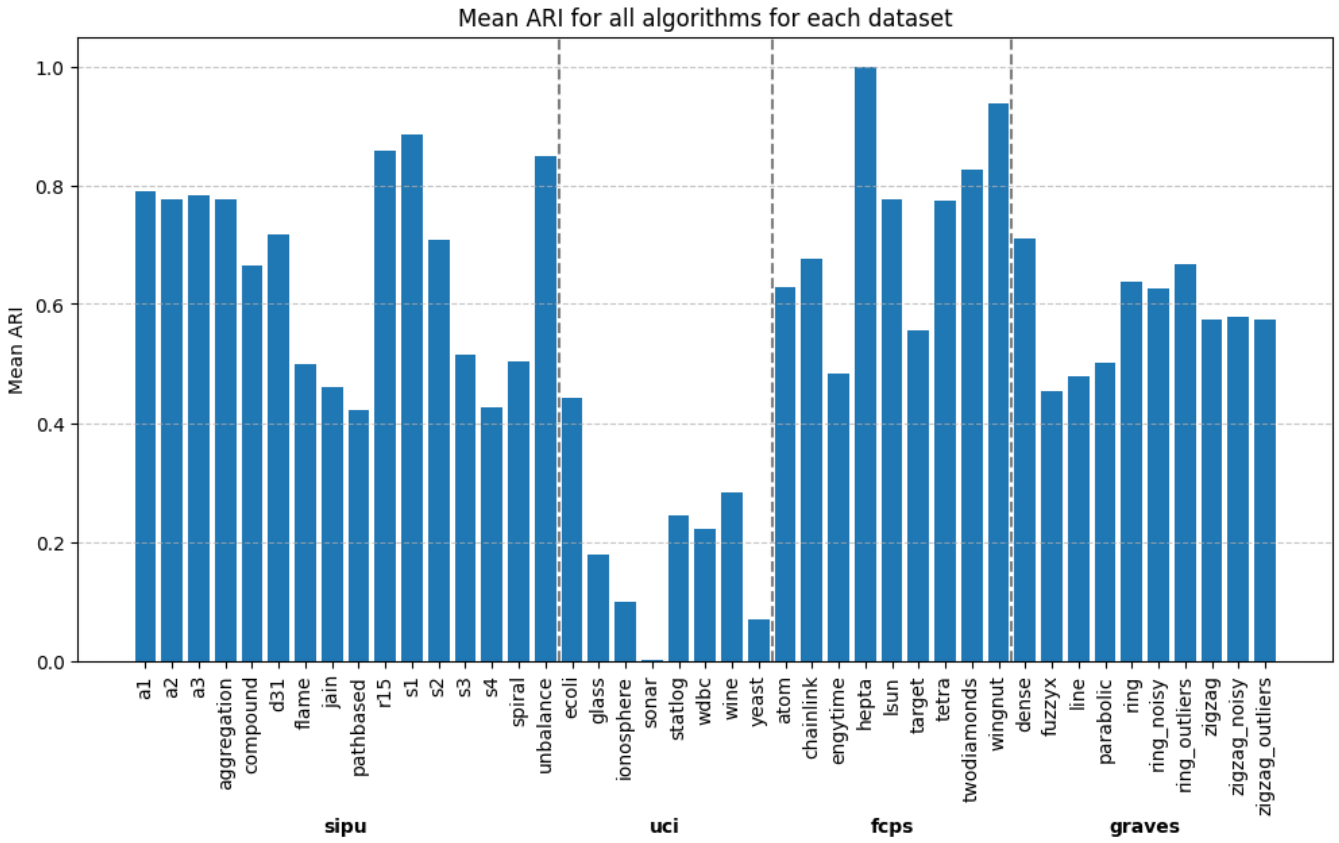
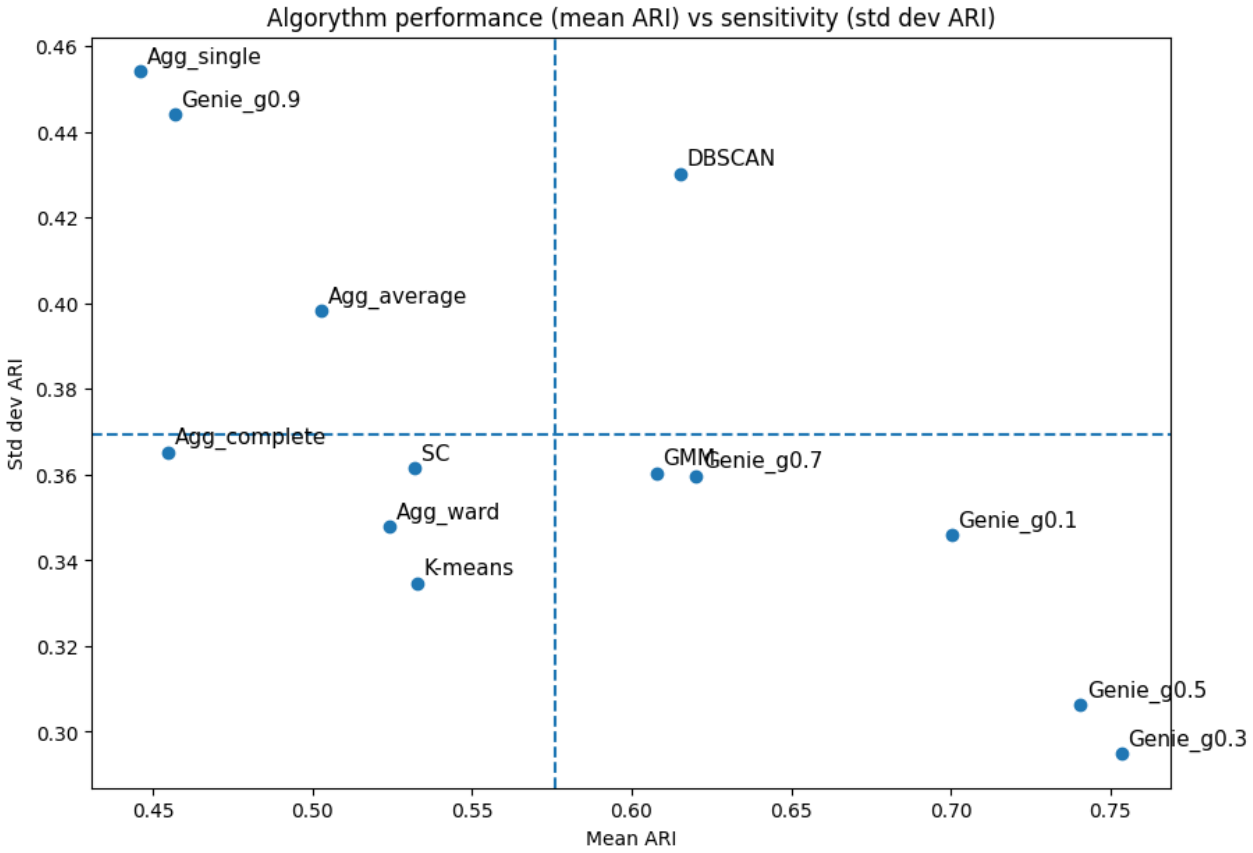


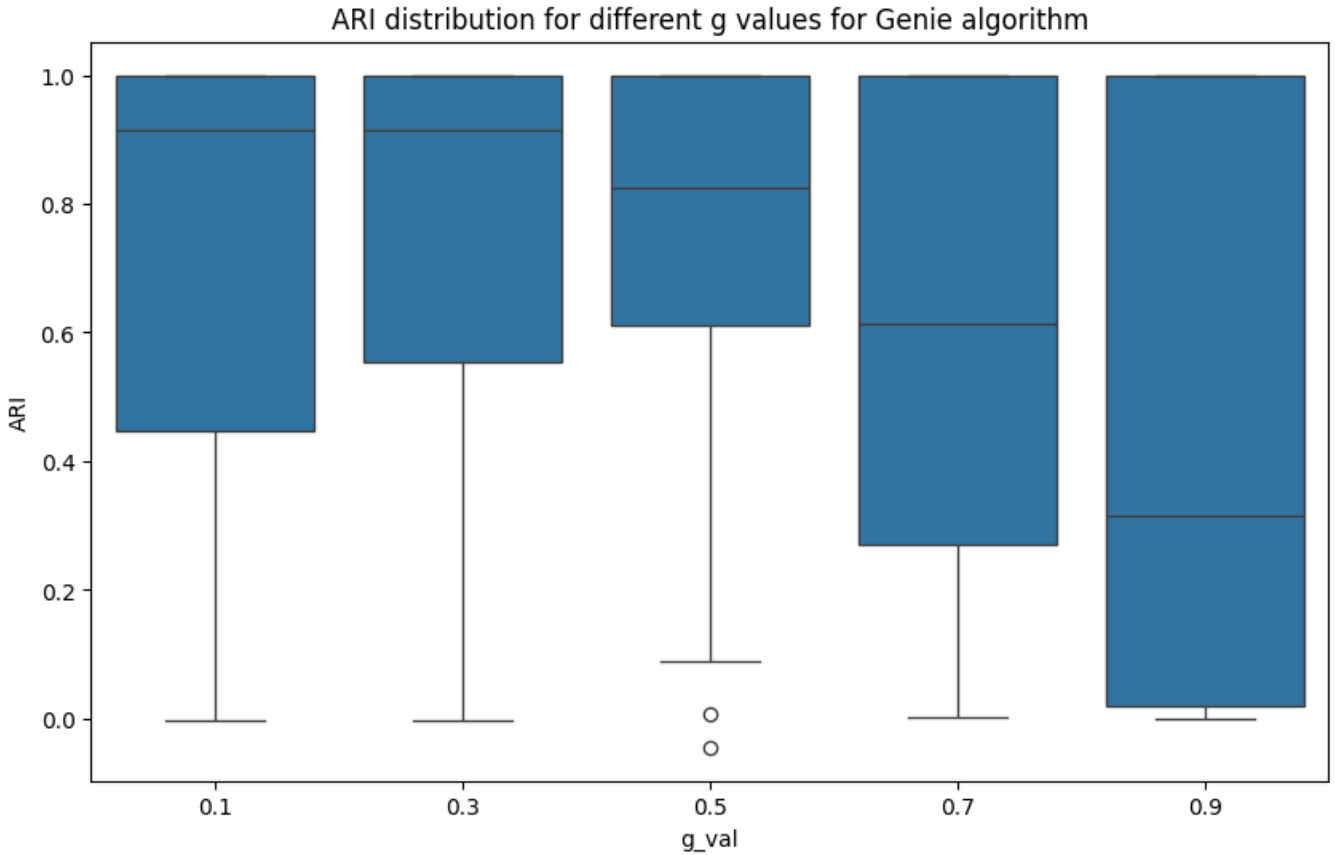
Figure: Mean ARI achieved by all algorithms dor each of the datasets, with respect to their battery.The uci battery was the most difficult on average.

Algorithm performance and sensitivity

This visualisation represents the accuracy and stability of the clustering algorithms. The genie model, with different g parameter values, achieves high mean ARI, with low standard deviation as the same time.

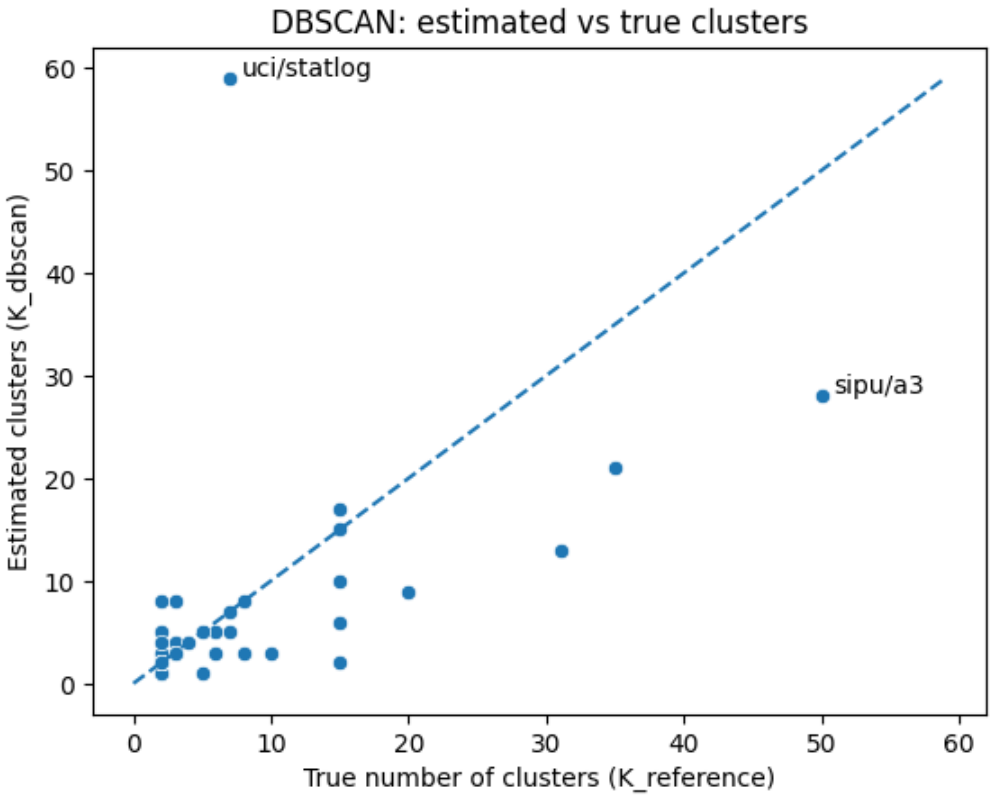


Genie parameter influence



The genie algorithm for $g = 0.3$ achieves not only the best average ARI value, but also has very stable performance across across all datasets.

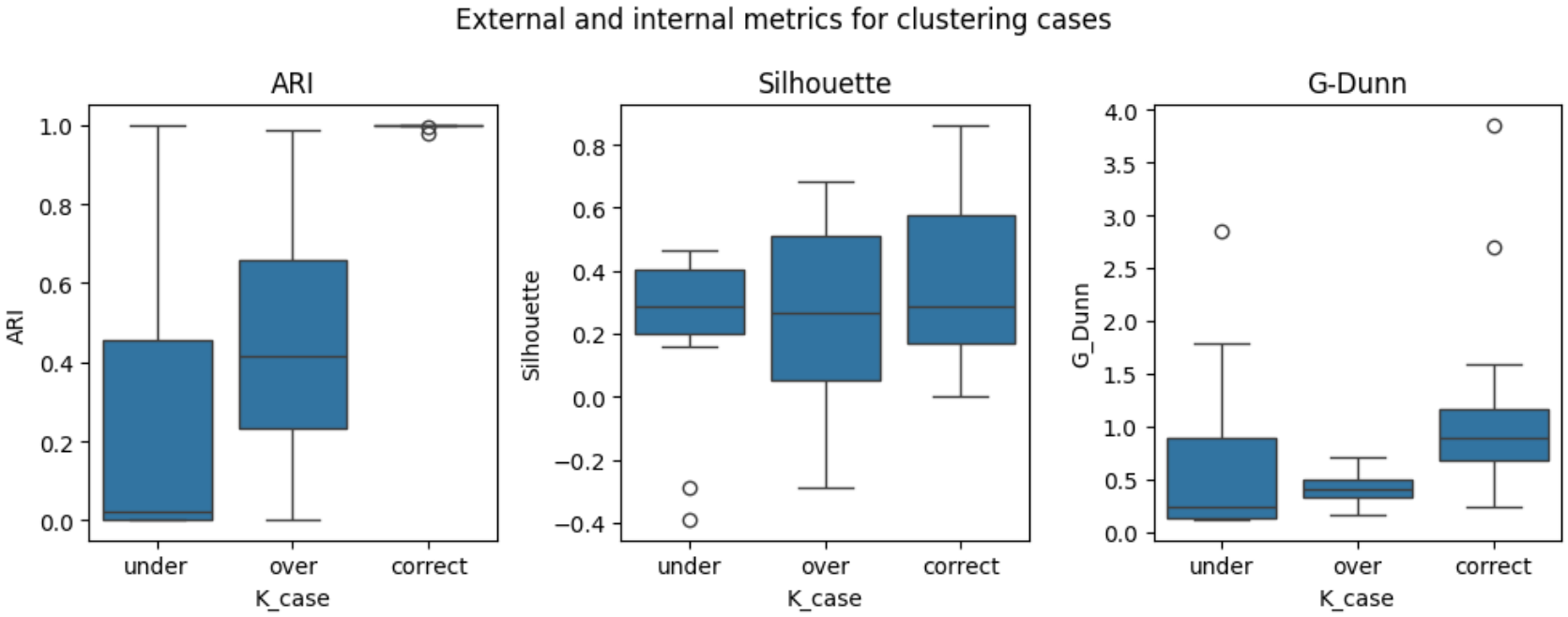
DBSCAN



K_case	Number of cases
correct	19
over	8
under	16

Finding the right number of clusters is a significant DBSCAN issue. The algorithms under- or overestimated the number of clusters for 56% of the datasets.

DBSCAN



Internal validation measures, especially Silhouette, which prefers spherical clusters, while DBSCAN can produce irregularly shaped ones, may indicate good clustering quality even when the clustering structure is incorrect.

Summary

1. The Genie algorithm achieves the best overall performance, combining high clustering accuracy with low variance across datasets, indicating strong robustness.
2. DBSCAN is hard to tune and fails often at cluster count estimation, however when it estimates the correct number of clusters its performance is extremely strong.
3. Internal validation metrics may be misleading when assessing clustering quality, as they do not always reflect agreement with "ground-truth" labels.

Thank you for your attention!